

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Program (Matlab Code)

```
clc; clear
```

```
Fs=10000; Fp=1000; Fst=1500;
```

```
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
```

```
[b,a]=butter(N,Wn);
```

```
freqz(b,a) % Magnitude & Phase
```

```
figure
```

```
[h,n]=impz(b,a,50);
```

```
...title('Impulse Response')
```

```
figure
```

```
stepz(b,a)
```

```
figure
```

```
zplane(b,a)
```

```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

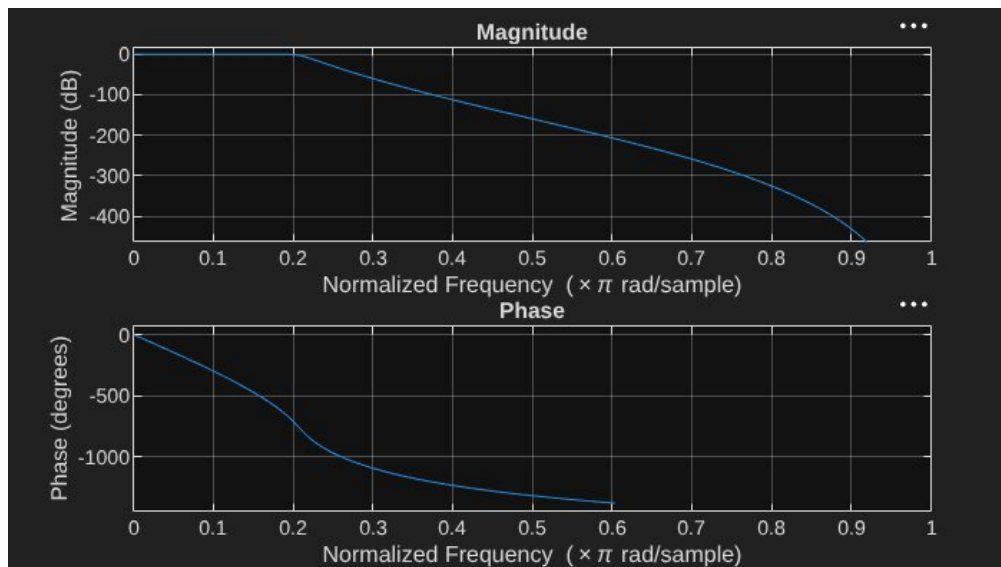
Output:

Objective 1:

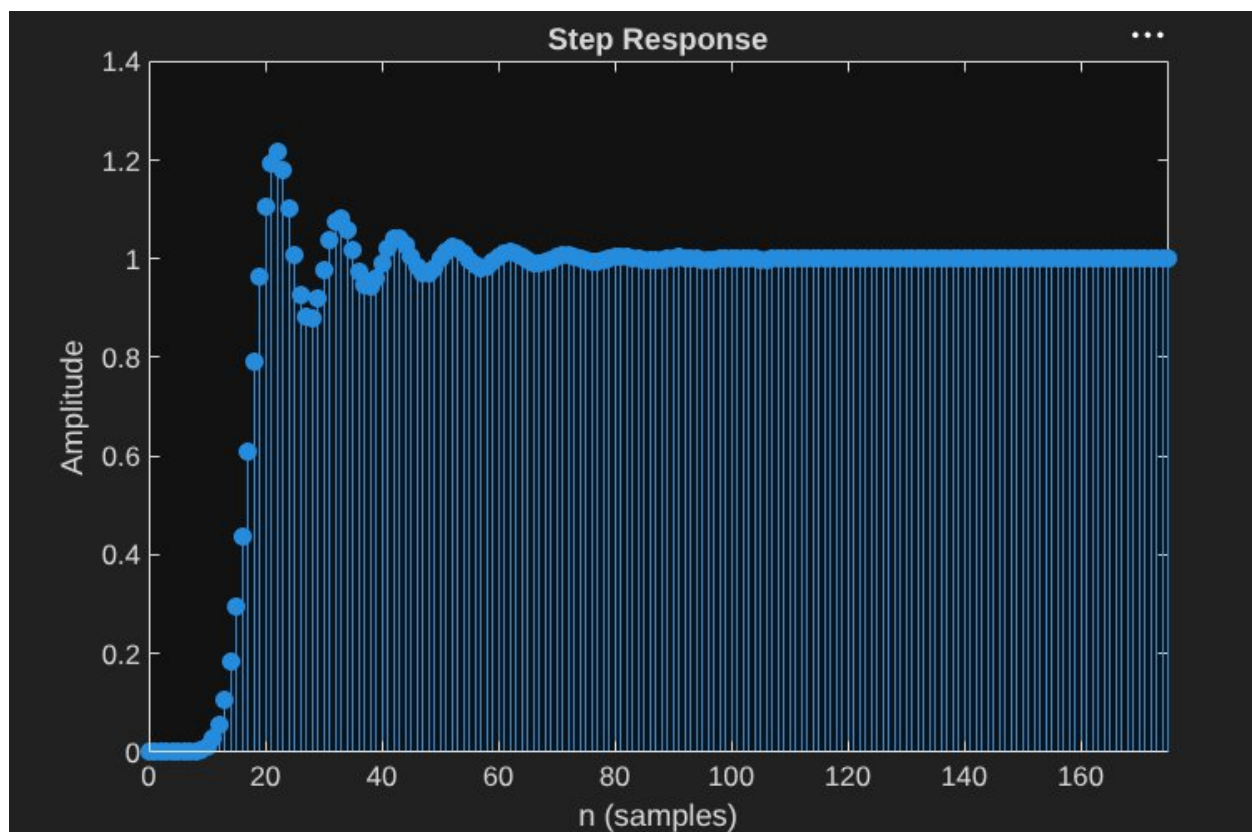
Minimum Filter Order (N) = 17

Cutoff Frequency (Wn) = 0.2083

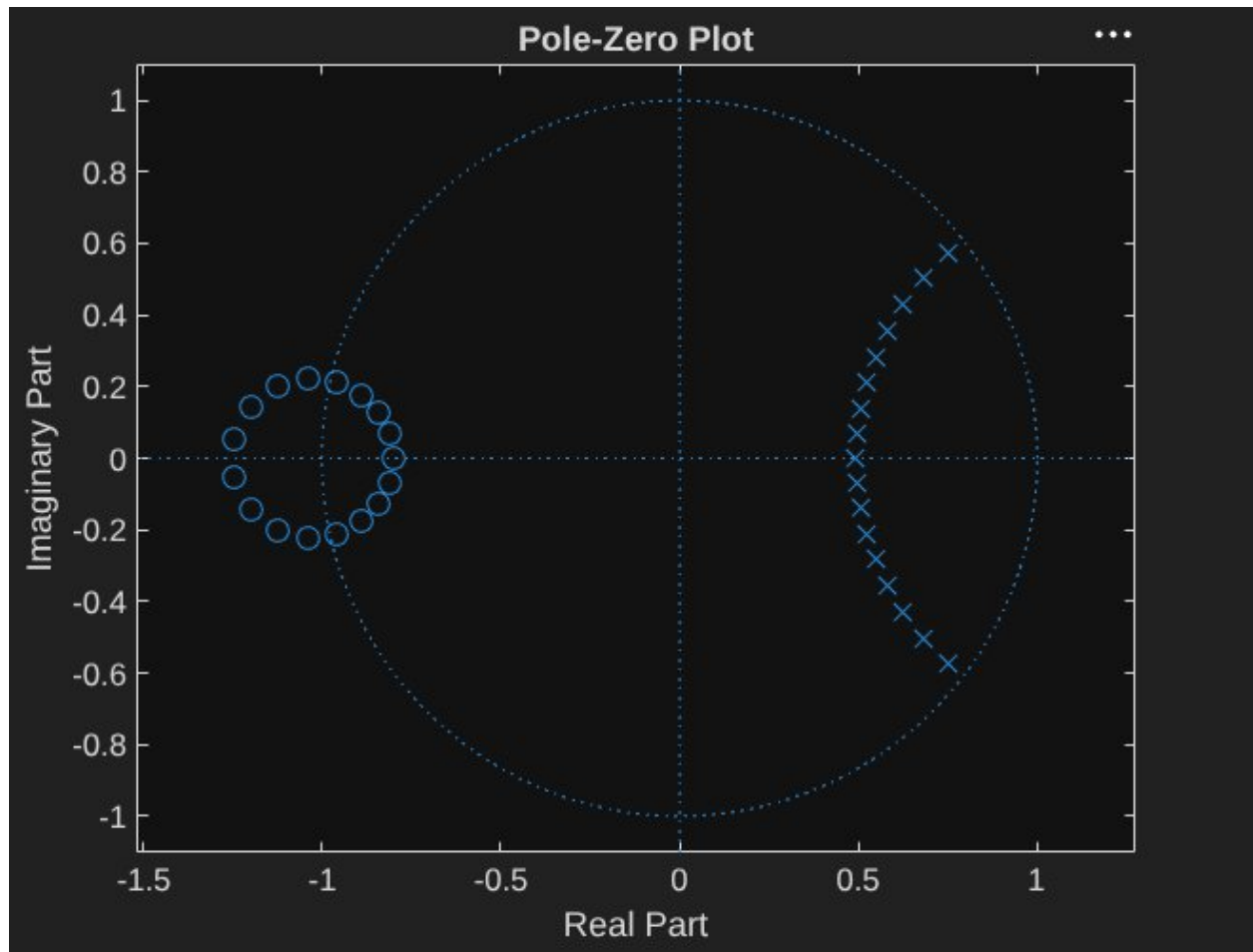
Objective 2:



Objective 3:



Objective 4:



1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Program (Matlab Code)

```
clc; clear
```

EX NO:1

Butterworth IIR Filters

DATE:

1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```
Fs=10000; Fp=3000; Fst=2000;  
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);  
[b,a]=butter(N,Wn,'high');  
  
% Magnitude & Phase  
freqz(b,a)  
  
% Impulse Response  
figure  
[h,n]=impz(b,a,50);  
stem(n,h), grid on  
title('Impulse Response')  
  
% Step Response  
figure  
stepz(b,a)  
title('Step Response')  
  
% Group Delay  
figure  
grpdelay(b,a)  
title('Group Delay')  
  
% Pole-Zero Plot  
figure  
zplane(b,a)  
title('Pole-Zero Plot')
```

```
fprintf('Minimum Filter Order (N) = %d\n',N);  
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

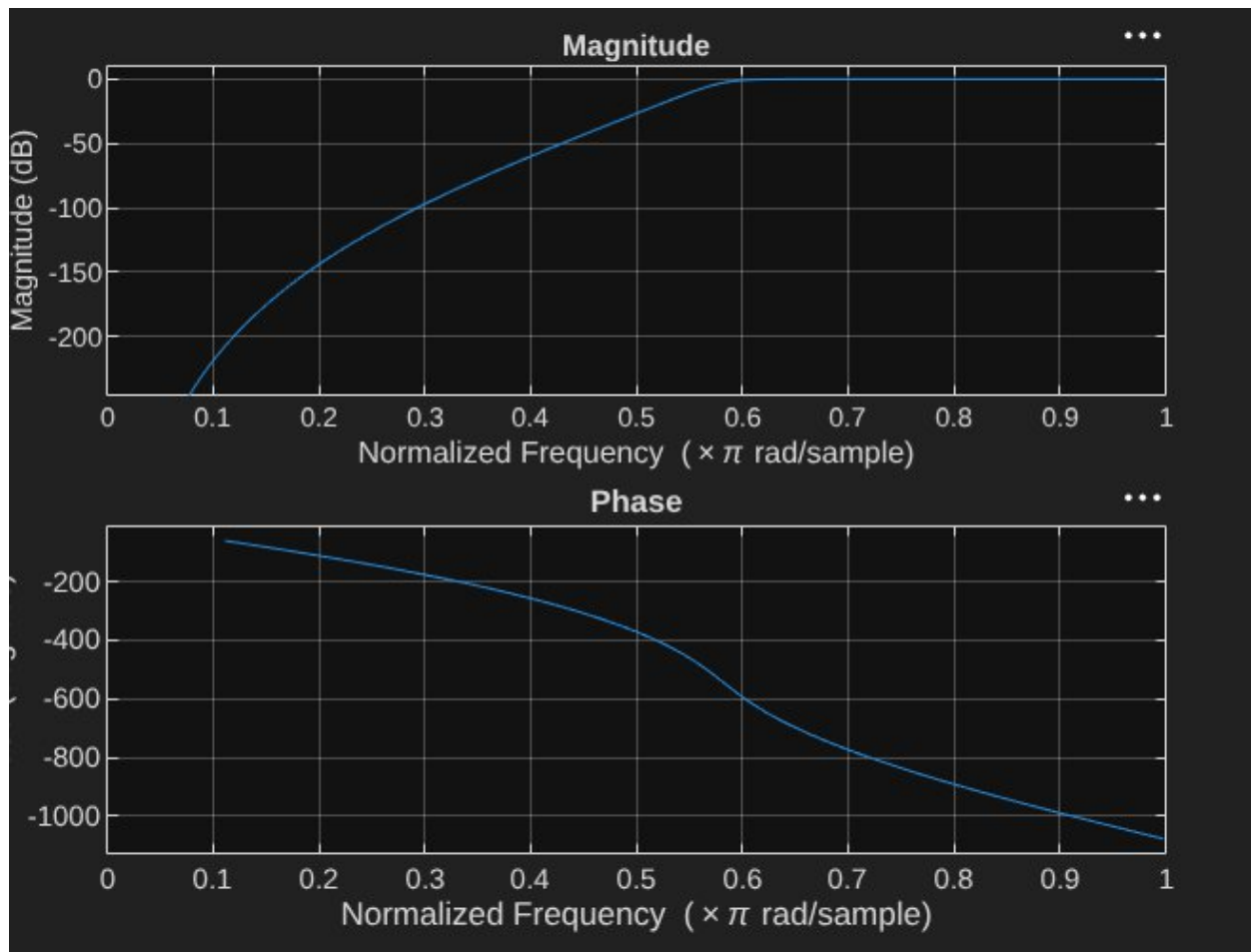
Output:

Objective 1:

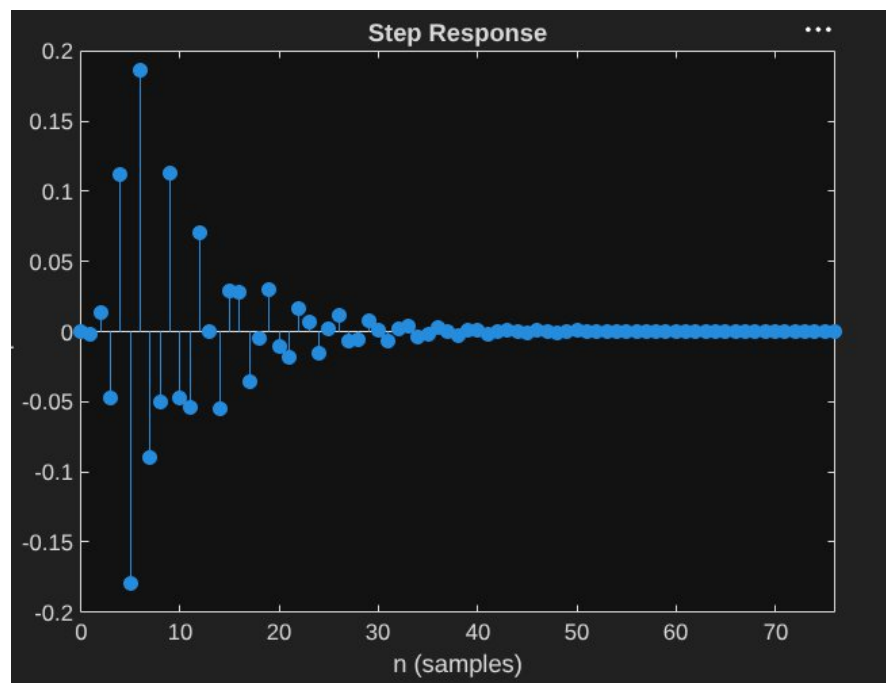
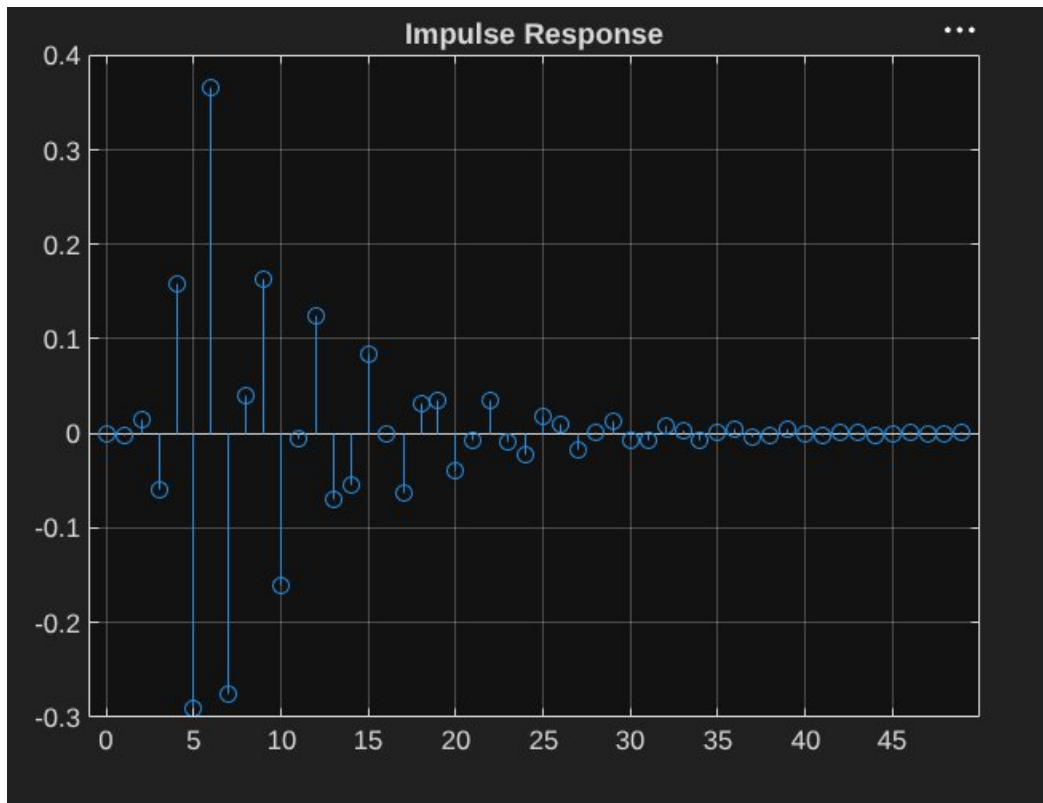
Minimum Filter Order (N) = 12

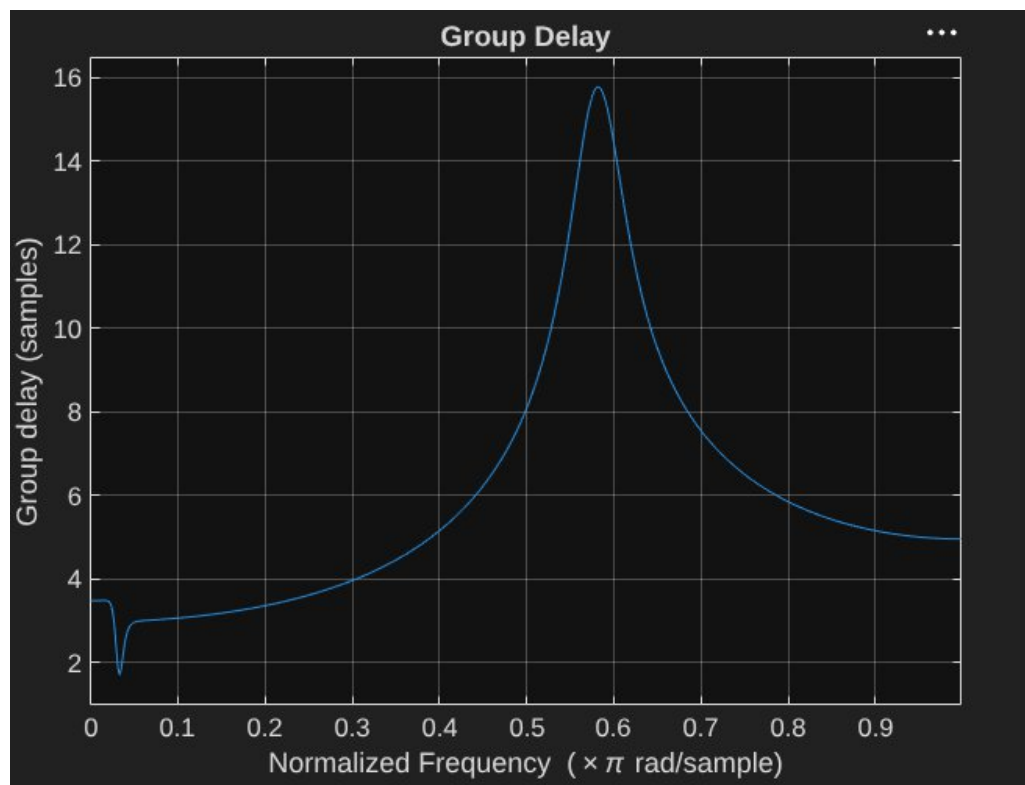
Cutoff Frequency (Wn) = 0.5807

Objective 2:



Objective 3:





Objective 4:

